INCLUSIVE MOBILE LEARNING APP FOR VISUALLY, HEARING, PHYSICALLY AND COGNITIVELY IMPAIRED STUDENTS (GRADE 3-5)

25-26J-207

Project Proposal Report

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Department of Information Technology

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1. DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Mathota Arachchi S.S.	IT22070876	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

……………………………. …………………

Signature of the supervisor Date

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Signature of the co-supervisor Date

2. ABSTRACT

Students in Sri Lanka with cognitive disabilities in kids with cognitive and learning difficulties (IQ-band adaptive: below / average / above average) face significant barriers in traditional classrooms due to standardized teaching methods, language barriers, and a lack of inclusive digital resources. While adaptive learning platforms and gamified systems exist internationally, they are rarely localized to the Sri Lankan context or tailored to different IQ levels. This proposal introduces a Cognitive & Adaptive Mobile Learning Platform, designed using Flutter, Firebase, and TensorFlow Lite, that delivers personalized, gamified, and multi-sensory learning activities in Sinhala. The system adapts content difficulty to the learner's IQ band (below average, average, above average) while providing animated character-based emotional guidance to maintain motivation and engagement.

The methodology employs a modular, iterative design that incorporates adaptive learning algorithms, gamified micro-activities, and culturally relevant content. Real-world pilot testing will be conducted with students, supported by caregivers and educators, to evaluate effectiveness, usability, and engagement. The platform emphasizes accessibility through universal design principles, offline support, and progress-tracking dashboards for educators. With strong commercialization potential, the solution addresses a critical gap in inclusive education, offering a scalable pathway to enhance academic outcomes and social inclusion for students with cognitive disabilities in Sri Lanka.

Keywords: Cognitive & Adaptive Mobile Learning Platform, Cognitive disabilities, Personalized learning, Gamification.

3. TABLE OF CONTENTS

4. LIST OF FIGURES

5. LIST OF TABLES

6. LIST OF ABBREVIATIONS

- AI: Artificial Intelligence
- API: Application Programming Interface
- ADHD: Attention Deficit Hyperactivity Disorder
- HTTP: Hyper Text Transfer Protocol
- IQ: Intelligence Quotient
- NIE: National Institute of Education
- SDK: Software Development Kit
- TLS: Transport Layer Security
- UDL: Universal Design for Learning
- UI: User Interface
- WCAG: Web Content Accessibility Guidelines

7. LIST OF APPENDICES

8. INTRODUCTION

8.1 Background

Cognitive disabilities affect children's ability to learn, remember, and apply knowledge in classroom environments. In Sri Lanka, many primary school students with cognitive and learning difficulties experience challenges such as slower processing, weak memory retention, short attention spans, and difficulty with abstract reasoning [1], [6]. Traditional teaching practices, which rely heavily on uniform pacing and abstract explanations, often fail to meet these diverse needs, resulting in disengagement, low performance, and reduced inclusion.

Globally, about 2–3% of school-aged children in developing countries live with cognitive disabilities [2]. Within Sri Lanka, the standardized education system offers limited flexibility, which restricts opportunities for individualized learning and contributes to low self-esteem among these learners. Although specialized interventions exist, fewer than 5% of children who require support are able to access them, leaving a significant portion of students underserved.

Research has demonstrated that adaptive learning, gamified lessons, and visual aids can enhance learning outcomes for students with cognitive disabilities [3]. Personalized pacing, interactive activities, and real-time progress tracking help address diverse learning needs while maintaining motivation. However, most of these tools are designed for English-speaking contexts and are rarely tailored for Sinhala-speaking students.

Mobile learning applications supported by artificial intelligence (AI) and adaptive algorithms provide an opportunity to deliver personalized instruction, portability, and multimodal support [4]. These systems can adjust lesson difficulty in real time, match learning to individual cognitive profiles, and provide both academic and emotional support. For example, AI-powered adaptive frameworks have been proposed to align instructional complexity with learners' IQ levels, ensuring that lessons are neither overwhelming nor under stimulating [5].

Despite these advances, students in Sri Lanka continue to face barriers due to the lack of localized, inclusive, and curriculum-aligned platforms. Cognitive disabilities remain a critical challenge to equitable education, underscoring the need for an adaptive and culturally relevant mobile learning solution.

8.2 Literature Review

Adaptive Learning Systems for Intellectual Disabilities in Developing Countries [1]

This study examined adaptive learning systems for children with intellectual disabilities in developing countries. The authors showed that adaptive content sequencing and personalized feedback significantly improved academic engagement and retention compared to traditional classroom methods. The work demonstrates that low-cost adaptive solutions can address systemic limitations in resource-constrained contexts, which parallels the situation in Sri Lanka. For this project, it provides evidence that adaptive technologies can deliver measurable gains even when teaching resources are scarce, supporting the need for a mobile, locally adapted platform.

Prevalence and Classroom Challenges of Cognitive Disabilities in Sri Lanka [2]

Focusing on Sri Lankan classrooms, this study mapped the prevalence of cognitive disabilities and documented common barriers such as memory retention difficulties, low attention, and poor comprehension under rigid teaching practices. The findings confirm that less than 5% of affected students receive targeted interventions. This local perspective strongly justifies your proposal, as it highlights the gap between student needs and the limited inclusive infrastructure available in schools, underscoring the urgency of a Sinhala-first adaptive learning platform.

Adaptive Learning Technologies: Systematic Review and Meta-Analysis [3]

Through a systematic review and meta-analysis, the authors investigated adaptive technologies for learners with cognitive impairments. They found that tailored pacing, multimodal learning, and responsive feedback systems improved learning outcomes across different age groups. Importantly, the analysis highlighted the benefits of combining gamification with adaptation, making lessons shorter and more engaging. This directly informs your project's design of gamified micro-activities aligned with IQ bands to enhance cognitive development.

Technology-Enhanced Learning Environments [4]

This paper presented results from randomized controlled trials on technology-enhanced learning environments for students with cognitive disabilities. Evidence showed that structured digital interventions improved memory, processing speed, and classroom participation. The use of rigorous RCTs strengthens the credibility of digital solutions in education. For your project, it emphasizes the importance of testing effectiveness through pilot studies with clear outcome metrics such as accuracy, engagement, and retention.

AI-Driven Personalized Learning Pathways for Students with Disabilities [5]

The authors explored AI-driven personalized learning pathways for children with learning disabilities. Their framework automatically adjusted lesson difficulty and content order based on learner profiles and real-time responses. This approach significantly improved motivation and reduced frustration. The relevance to your study is direct, as your proposal aims to use IQ-band adaptation and dynamic difficulty scaling, ensuring that lessons are neither too easy nor overwhelming.

Cognitive Processing Challenges in Sri Lankan Primary Education [6]

This research focused on cognitive processing challenges in Sri Lankan classrooms, including difficulties in abstract reasoning, slow recall, and reduced attention. Teachers reported struggling to adapt lessons to diverse learner needs due to lack of training and tools. These findings explain why conventional instruction fails many students locally.

They provide critical justification for your emphasis on concrete examples, short tasks, and simplified digital interfaces in the proposed mobile app.

Inclusive Pedagogy Framework for Educational Technology Design [7]

This work introduced an inclusive pedagogy framework for designing educational technology. It emphasized providing multiple means of representation, engagement, and expression so that students with disabilities are not excluded. The framework is highly relevant to your study, guiding the integration of audio, visuals, animations, and text in Sinhala. It ensures the mobile app follows universal design principles that enable all learners to participate meaningfully.

Machine Learning Approaches for Cognitive Load Optimization [8]

This survey reviewed machine learning approaches for adaptive educational systems, with a focus on optimizing cognitive load. The study stressed that balancing task difficulty and learner capacity prevents overload and increases retention. It provides theoretical support for using adaptive algorithms to modulate pacing and complexity. Your project builds on this by tailoring lesson difficulty to children's IQ levels and performance in real time, thus controlling cognitive load.

Game-Based Micro-Learning for Students with Attention Difficulties [9]

The authors designed and validated game-based micro-learning activities for students with attention difficulties. Findings showed that short, game-like lessons increased focus and reduced dropouts. The principle of breaking content into smaller units directly supports your project's micro-activity design. It confirms that gamification is not just motivating but also effective in sustaining learning for children with limited attention spans.

Gamification in Special Education: Evidence-Based Approaches [10]

This research reviewed gamification practices in special education, highlighting how points, badges, and leaderboards can boost motivation and reinforce learning in students with cognitive disabilities. The study noted that gamification is most effective when linked to meaningful progress and feedback rather than just entertainment. This aligns with your proposed system's use of rewards, badges, and progress tracking to maintain engagement while reinforcing curriculum content.

Animated Pedagogical Agents in Adaptive Learning Systems [11]

Baylor and Kim investigated the use of animated pedagogical agents in adaptive learning systems and reported improvements in motivation, emotional regulation, and learning outcomes among students with cognitive challenges. Their findings suggest that animated feedback can reduce frustration and encourage persistence. This evidence supports the integration of animated emotional guidance in the proposed Sinhala-based system, providing both academic and emotional benefits for learners.

Cultural Localization in South Asian Educational Technology [12]

This research examined cultural localization in South Asian ed-tech platforms. It concluded that culturally relevant content and local language support significantly increase acceptance and effectiveness. Generic, Western-oriented tools often fail in rural Asian settings. This is highly relevant to your project, as you are prioritizing Sinhala-first design and curriculum alignment, ensuring the app resonates with local learners and teachers.

Universal Design for Learning with Mobile Classroom Technology [13]

The authors applied Universal Design for Learning (UDL) to mobile classroom technology. Their framework included flexible learning pathways, accessible assessments, and multi-modal supports. Results showed that UDL increases participation of students with disabilities in mainstream classrooms. This directly informs your system's flexible activity design, child-friendly interfaces, and multiple modes of learning support (text, voice, visuals).

AI-Driven Experiments in Adaptive Learning for Special Education [14]

This paper presented AI-driven adaptive learning systems applied to special education. It emphasized iterative experimentation and data analytics to refine interventions. The study suggests that constant monitoring and adjustment improve effectiveness. For your project, this supports including progress dashboards and analytics for teachers, allowing iterative refinement of learning activities.

Digital Divide Adoption in Sri Lankan Special Education [15]

The authors discussed the digital divide in Sri Lankan special education, noting barriers such as poor infrastructure, limited training, and affordability issues. They recommended low-cost, offline-enabled solutions. This is directly relevant to your platform, which supports offline mode, low-spec devices, and simple onboarding. It ensures accessibility even in rural schools where connectivity is limited.

Flexible Collaborative Scripts in Multicultural Learning Contexts [16]

This study analyzed flexible collaborative scripts in multicultural educational contexts. It highlighted those rigid systems often fail in diverse environments and that adaptability to cultural and linguistic diversity is key. Your proposal reflects this principle by embedding Sinhala content, adapting tasks to IQ levels, and considering Sri Lankan pedagogy, making lessons contextually appropriate.

Accessibility Design Factors in Blended Learning Environments [17]

The study reviewed design factors for accessibility in blended learning environments. It found that navigation clarity, consistent feedback, and simplified interaction design improved accessibility for learners with disabilities. These insights guide your app's child-friendly interface, large buttons, simplified layouts, and clear interaction flows, ensuring usability for young learners with cognitive difficulties.

Inclusive Education Policies and Implementation in South Asia [18]

This research analyzed inclusive education policy frameworks in South Asia, finding gaps between policy and classroom implementation. While policies promote inclusion, schools lack tools and training to operationalize them. This highlights the value of your mobile platform as a practical, scalable tool that can help implement inclusive policies at the ground level.

Layout and Readability Guidelines for Dyslexia and Cognitive Needs [19]

The authors proposed layout and readability guidelines to improve accessibility for students with dyslexia and cognitive impairments. Their web service demonstrated that font size, spacing, and color contrast directly affect comprehension. These findings influence your design, where Sinhala typography, spacing, and child-friendly visuals must follow accessibility standards to reduce cognitive strain.

Cognitive Telerehabilitation During COVID-19: A Pilot Study [20]

This pilot study explored cognitive telerehabilitation for adolescents with learning disabilities during COVID-19 lockdowns. The results showed that structured digital training-maintained learning progress despite school closures. This confirms the feasibility of remote interventions and supports your design of a mobile tool usable both in classrooms and at home with caregiver supervision.

Effects of Cognitive Training Programs on School-Age Children [21]

Looney's research evaluated cognitive training programs for school-aged children and found improvements in working memory, attention, and processing speed. The results support the integration of targeted memory and attention-building exercises into digital platforms. Your project builds on this by embedding such exercises within gamified micro-learning modules.

Scaffolding and Inclusive Co-Design for Cognitive Disabilities [22]

This work introduced scaffolding approaches for inclusive co-design with learners who have cognitive disabilities. It emphasized involving learners and caregivers in the design process and gradually reducing support as skills improve. Your project can adopt this by embedding hints, guided steps, and teacher/caregiver collaboration, ensuring that learners are supported but also encouraged to become independent.

Assistive Technologies for Dyslexia: Multisensory Review (2015–2023) [23]

Paudel reviewed assistive technologies for dyslexia, noting growth in multisensory tools (AR, VR, haptics). However, the review highlighted challenges of limited personalization, cost, and poor scalability in low-resource contexts. These findings align with your project's focus on lightweight, personalized, and locally sustainable solutions that avoid over-complexity while ensuring accessibility.

AI-Based Psychometric Systems for Adaptive Tutoring [24]

Hu proposed an AI-based psychometric system for adaptive tutoring, linking cognitive assessments with personalized lesson delivery. The framework ensures that difficulty matches the learner's broader cognitive profile, not just past performance. Your project takes inspiration from this approach by including IQ-band assessments and adaptive adjustments, allowing activities to be matched to each child's cognitive level.

8.3 Research Gap

Global research has explored mobile learning, gamification, and adaptive education for students with cognitive disabilities. However, there is still no solution that combines IQ-specific adaptation, animated emotional guidance, and Sinhala-language localization for Sri Lankan primary students with cognitive and learning difficulties. A review of assistive technologies for dyslexia showed a rapid increase in multisensory tools such as AR, VR, and haptics, but also noted issues of limited personalization, short trial periods, and poor scalability in low-resource settings [22]. These gaps highlight the need for culturally adapted, sustainable, and personalized interventions for cognitive learners in Sri Lanka.

8.3.1. Identified gaps in the literature

- Cultural and Linguistic Localization

  Most tools are designed for English-speaking or Western contexts, with limited alignment to the Sri Lankan curriculum or cultural environment. They emphasize generic skills rather than personalized, curriculum-relevant activities across different IQ bands (below average, average, above average) [16].

- Integration of Pedagogical Approaches

  Existing systems typically address isolated features such as adaptive pacing or gamification. Few platforms integrate real-time adaptation, animated character guidance, and multisensory micro-activities that also provide emotional support tailored to learners with short attention spans.

- Accessibility and Usability

  Mainstream e-learning tools, such as Google Classroom or Khan Academy, though content-rich, are not optimized for independent use by children with cognitive disabilities. Interfaces often lack simplified navigation, consistent interaction, and culturally relevant storytelling [17].

- Absence of Comprehensive Solutions

  Currently, no single platform offers:

    - IQ-specific adaptive micro-activities aligned with the Sri Lankan curriculum.
    - Sinhala-language animated character guidance.
    - Gamified reinforcement with visual rewards.
    - Multisensory integration (text, sound, images, animations).
    - Child-friendly accessibility features for cognitive learners.

- Evidence-Based Design

  Many applications are developed without rigorous validation through controlled studies, which limits their effectiveness and applicability to the target group.

- Scalability and Sustainability

  Research efforts often overlook resource constraints in countries like Sri Lanka, which reduces the long-term viability and adoption of these efforts.

Table 8.1: Comparison of Research Gaps in Existing Studies and Proposed Mobile Application

Feature / Research	Sinhala language	IQ-specific adaptation	Gamification / micro-learning	Animated emotional guidance	Progress tracking & reports
Proposed System	✓	✓	✓	✓	✓
Anderson et al (2022) [4]	✗	PARTIALLY	PARTIALLY	✗	✓
Kiili & Kiili (2022) [4]	✗	✗	PARTIALLY	✗	PARTIALLY
Prensky & Digital (2022) [10]	✗	✗	PARTIALLY	✗	PARTIALLY
Baylor & Kim (2022) [11]	✗	✗	PARTIALLY	✓	PARTIALLY
Hall, Meyer & Rose (2022) [13]	✗	✗	PARTIALLY	✗	✓
Maggio et al. (2021) [20]	✗	✗	PARTIALLY	✗	✓
Looney (2024) [21]	✗	✗	PARTIALLY	✗	✓
Paudel (2024) [23]	✗	✗	PARTIALLY	✗	✗

8.4 Research Problem

In Sri Lanka, many primary school students in kids with cognitive and learning difficulties with cognitive disabilities, including intellectual disabilities, ADHD, autism spectrum disorders, and specific learning difficulties, struggle to follow classroom lessons. Standardized teaching methods and fixed learning paces often overwhelm slower learners while failing to challenge others, leading to poor comprehension, slow progress, and disengagement.

1. **Why is there a need to create a mobile learning app for students with cognitive and learning difficulties in Sri Lanka?**

   Because many primary school students with intellectual disabilities, ADHD, autism, or specific learning difficulties struggle to keep up in traditional classrooms. Standard teaching methods move too fast for some and too slow for others, leading to poor comprehension, frustration, and disengagement.

2. **What limitations exist in the current education system that make this app necessary?**

   Sri Lankan classrooms mostly rely on chalk-and-talk teaching. Teachers often lack training in inclusive education, and fewer than 5% of students who need special support receive it. This creates a wide gap in equal learning opportunities.

3. **Why are existing international tools not sufficient for these learners?**

   Most global tools are designed for neurotypical students and focus mainly on visuals or adaptive testing. They do not combine emotional guidance, personalized pacing, and Sinhala-language support. As a result, they are not culturally relevant or usable for many Sri Lankan students.

4. **Why is it important to introduce multi-modal and gamified learning?**

   Children with cognitive difficulties learn better through multiple senses (hearing, seeing, touching, and interacting). Combining text, audio, images, animations, and games helps maintain attention, improve memory, and keep students motivated. Current tools rarely offer this in one platform.

5. **What will happen if such an app is not developed?**

   Without a localized, adaptive, and supportive tool, many Sri Lankan students with cognitive and learning difficulties will continue to fall behind academically. This not only reduces their personal educational outcomes but also limits national goals of inclusion and social development.

9. OBJECTIVES

9.1 Main Objectives

The aim is to design, develop, and evaluate a mobile-based Cognitive & Adaptive Learning Platform using Flutter. The system will deliver personalized, engaging, and multi-sensory educational activities for Sri Lankan primary school students with cognitive and learning difficulties. Lessons will adapt to different IQ bands (below average, average, above average) while providing emotional guidance and accessibility features to improve learning outcomes. Building on this, researchers have proposed AI-based psychometric tutoring frameworks that link cognitive assessments with adaptive content delivery [24]. Inspired by these ideas, this project will apply lightweight psychometric methods to ensure that content difficulty matches both learner performance and broader cognitive profiles.

9.2 Specific Objectives

- Conduct a multidisciplinary assessment of needs with teachers, psychologists, and caregivers to identify learning barriers and targeted support strategies that are required.
- Develop an adaptive learning system using machine learning to adapt content, pace, and instruction based on a student's IQ and existing performance.
- Develop gamified micro-learning modules to sustain learner engagement and build cognitive skills through evidence-based pedagogy and game mechanics.

- Embark on the deployment of animated emotional guidance agents to provide encouragement, corrective feedback, and supportive interventions at moments of frustration or disengagement.
- Introduce multimodal accessibility features, such as Sinhala language support, voice guidance, and visual cues, aligned with global accessibility standards.
- Implement an AI-based assessment module offering interactive quizzes with prompt feedback, tracking of progress, and rewards for achievement for formative and summative assessment.

10.METHODOLOGY

The proposed research focuses on developing a Cognitive & Adaptive Mobile Learning Platform for kids with cognitive and learning difficulties. The aim is to overcome barriers in traditional classrooms, such as standardized lesson pacing, abstract teaching methods, and a lack of motivational support, by providing personalized, gamified, and emotionally guided learning experiences in Sinhala. The design ensures inclusivity, accessibility, and alignment with the Sri Lankan primary school curriculum.

10.1System Architecture

The system follows a modular client-server architecture, enabling real-time adaptation, emotional support, and secure data management.

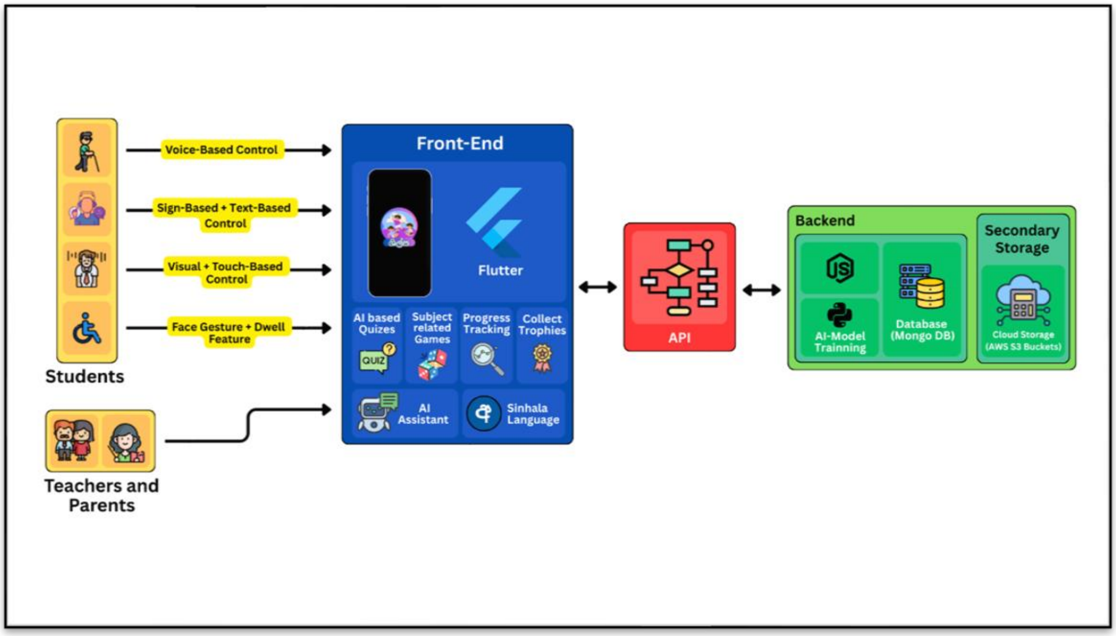

Fig.10.1: System overview diagram of the complete inclusive mobile learning app

Key Architecture Components:

- The mobile frontend, built with Flutter, delivers interactive micro-learning activities with animated character support and progress visualizations, using high-contrast colours, large touch targets, and simplified navigation to ensure accessibility.
- The adaptive learning engine applies AI and machine learning to adjust difficulty, sequencing, and pacing based on each learner's IQ level and real-

time performance, using methods such as reinforcement learning and natural language processing.

- The emotional guidance module detects errors, retries, or disengagement and responds with motivational animations and voice prompts, using sentiment analysis and behaviour recognition to provide appropriate support.
- The backend, developed with Firebase and Node.js, manages authentication, data storage, and analytics while ensuring scalability and reliability through backup and recovery mechanisms.
- The database, implemented with MongoDB or Firebase Realtime DB, securely stores learner profiles, IQ classifications, and performance data while supporting efficient analysis.
- The analytics and reporting system generate dashboards for caregivers and educators, offering real-time monitoring and historical trend analysis with personalized learning recommendations.

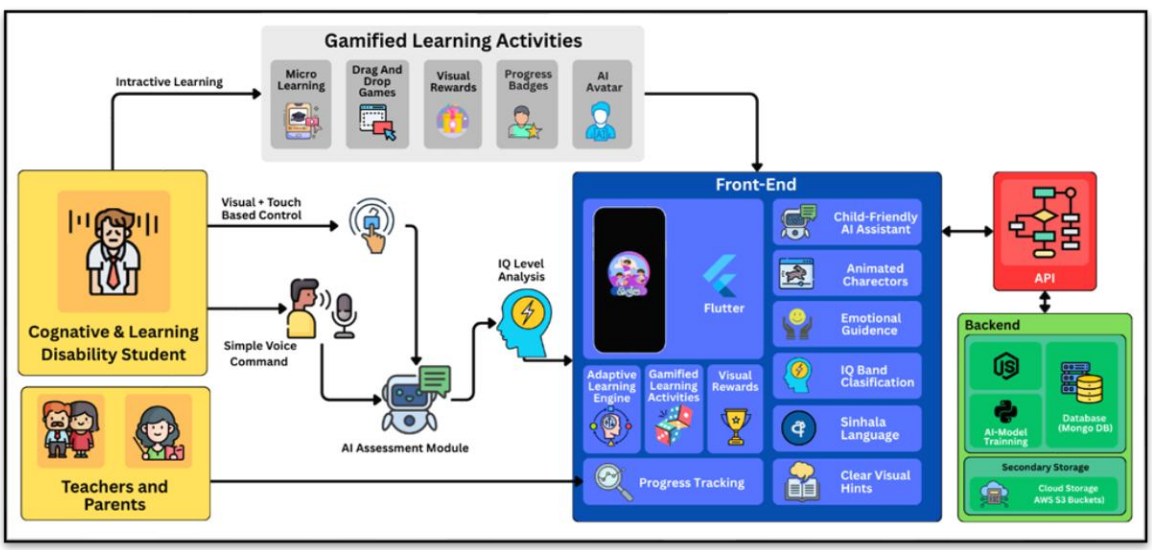

Fig. 10.2: System architecture overview for cognitive disability support

10.2Development Tools & Technologies

- The mobile application is developed using the Flutter SDK, which enables cross-platform deployment while providing high performance, a rich widget library, and built-in accessibility features.
- The backend relies on Node.js and Firebase to handle authentication, cloud storage, hosting, and API management, while also offering real-time database support essential for adaptive learning.
- Adaptive functionality is powered by TensorFlow Lite, which executes machine learning algorithms on-device to reduce latency, enable offline use, and provide personalized learning experiences.
- Gamification elements such as progress tracking, reward systems, and achievement badges are integrated into the learning modules, designed in alignment with motivational psychology and established game design principles.
- User interfaces are prototyped in Figma, emphasizing child-friendly layouts with large icons, minimal text, and Sinhala language integration, while following the Universal Design for Learning (UDL) guidelines.
- Accessibility is ensured by incorporating WCAG 2.1 standards alongside UDL principles, enabling the platform to be inclusive for learners with a wide range of disabilities.

10.3 Functional Modules & Workflow

1. User Registration & Profiles - Students, caregivers, and teachers create accounts; student profiles store IQ, learning preferences, and past performance.
2. Adaptive Content Delivery - Lessons are delivered as micro-activities that adjust in real time based on performance and learning style.
3. Animated Emotional Guidance - Context-sensitive encouragement and feedback are provided via animated characters.
4. Gamified Micro-Learning - Interactive games reinforce concepts and provide rewards for motivation.
5. Progress Tracking & Reports - Dashboards allow parents and educators to monitor progress and identify potential challenges.
6. Multi-Modal Learning Support - Content is delivered through text, narration, visuals, and interactive elements.
7. Offline Mode & Synchronisation - Core features work offline, with automatic syncing when online.

10.4 Data Collection Strategy

To evaluate the proposed system, data will be gathered from both primary and secondary sources, supported by pilot testing and ethical safeguards.

10.4.1. Primary data

- Observations**:** Classroom behavior and learning patterns of target students
- Interviews**:** Insights from teachers, parents, and educational psychologists
- Performance Data**:** Metrics such as completion rates, accuracy, engagement time, and progression
- Usability Testing**:** Formal sessions with students to assess ease of use

10.4.2. Secondary data

- Curriculum Review**:** Analysis of Sri Lankan cognitive students’ requirements.
- Literature Review**:** Existing studies on adaptive learning and assistive technologies.
- Comparative Review**:** Evaluation of current educational technology solutions.

10.5Expected Outcomes

At the completion of the project, the system will provide:

- Personalized, IQ-band-specific learning activities aligned with Sri Lankan curriculum standards.
- Animated emotional guidance that sustains motivation and reduces learning-related frustration.
- Comprehensive gamified reinforcement with badges, progress tracking, and achievement rewards.
- Multi-modal learning through Sinhala text, visual animations, narration.
- Real-time progress monitoring for parents and educators with actionable insights.
- Cross-platform deployment on Android and iOS with robust offline capabilities.
- Evidence-based validation of effectiveness through controlled pilot studies.
- Scalable architecture capable of supporting widespread deployment.
- Comprehensive documentation and training materials for stakeholders.

10.6 Disability-Specific Feature Mapping

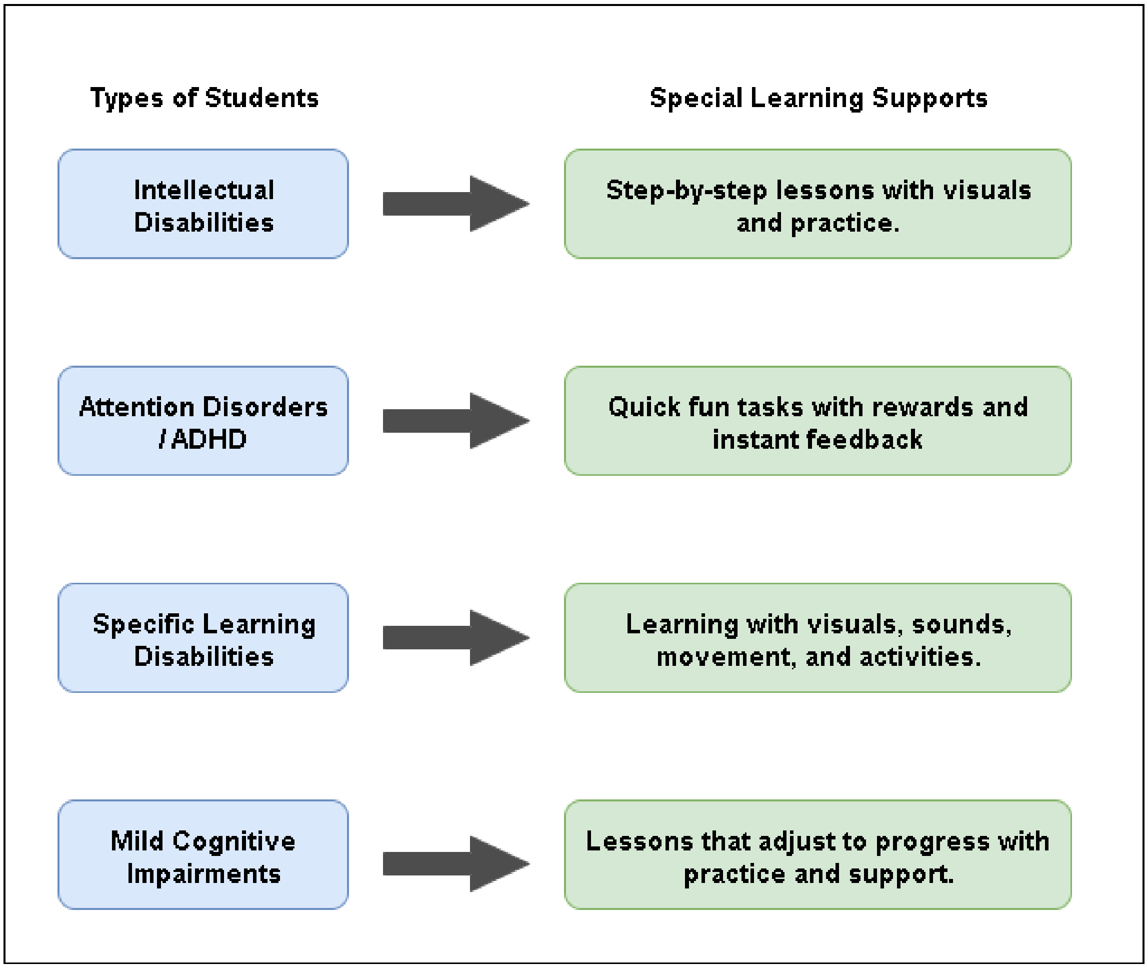

Fig. 10.3: Proposed system feature support mapped to different cognitive and learning disability types.

11.PROJECT REQUIREMENTS

11.1Functional Requirements

The system must support data-driven evaluation and inclusive learning through the following functions:

- Monitor student behavior and learning patterns during classroom use
- Collect structured feedback from teachers, parents, and psychologists.
- Record performance metrics such as task completion, accuracy, engagement time, and progress.
- Assess usability for students with cognitive disabilities.
- Align lessons and activities with Sri Lankan primary curriculum topics, delivered according to the learner’s IQ band (below / average / above average).
- Incorporate evidence from adaptive learning and assistive technology research.
- Compare platform features with existing educational technologies.
- Conduct pre-assessments to identify skills and preferences.
- Deliver structured micro-activities during learning sessions.
- Evaluate outcomes, motivation, and satisfaction after system use.
- Track retention and engagement over time.
- Apply statistical methods and algorithm testing to performance data.
- Analyze feedback thematically and develop case studies to interpret behavior patterns.

11.2Non-Functional Requirements

- Child-friendly with large buttons, clear visuals, and simple navigation.
- Follows WCAG 2.1 AA, supports screen readers and voice control.
- Interfaces customisable for different visual or motor needs.
- Activities load within 2 seconds and run smoothly at 60fps.
- Efficient memory use for low-spec devices and longer battery life.

- Automated data backups and recovery.
- Cloud uptime 99.5% with offline access and later sync.
- Functions well under poor network conditions.
- Supports 10,000+ users with horizontal scaling.
- CDN for fast delivery and optimised education databases.
- Data encrypted with HTTPS/TLS 1.3+.
- Secure login with multi-factor authentication for admins.
- Regular security audits and privacy compliance.
- Modular design for easy updates and feature expansion.
- Well-documented APIs for integration.
- Automated tests covering 80%+ of core features, with version control and rollback.
- Runs on Android (API 21+) and iOS (12+).
- Adapts to different screen sizes and device features.
- Backwards compatible with older school devices in Sri Lanka.

11.3 User Requirements

The system supports three main user groups. Students, Caregivers/Educators, and Administrators.

- **Students (kids with cognitive and learning difficulties)**
  - Easy-to-use interface with simple navigation.
  - Animated characters for emotional support and guidance.
  - Short activities (5–15 minutes) that match attention levels.
  - Instant feedback with visuals and sounds for motivation.
  - Multiple supports: Sinhala text-to-speech, animations, and interactive features.
  - Difficulty levels that adapt to each student’s ability and IQ.

    - Safe environment with age-appropriate content.

- **Caregivers and Educators**
    - Tools to create/manage student profiles with IQ and learning details.
    - Real-time dashboards to track progress and outcomes.
    - Options to set goals and customize learning paths.
    - Detailed reports with strengths, weaknesses, and suggestions.
    - Alerts for performance changes or concerns.
    - Training resources to use the platform effectively.
    - Communication tools for collaboration with others.

- **System Administrators**
    - Monitoring tools for system performance and usage.
    - Secure account and role management with permissions.
    - Content management for uploading and updating lessons.
    - Maintenance tools for updates, backups, and optimization.
    - Analytics dashboard showing overall usage and effectiveness.

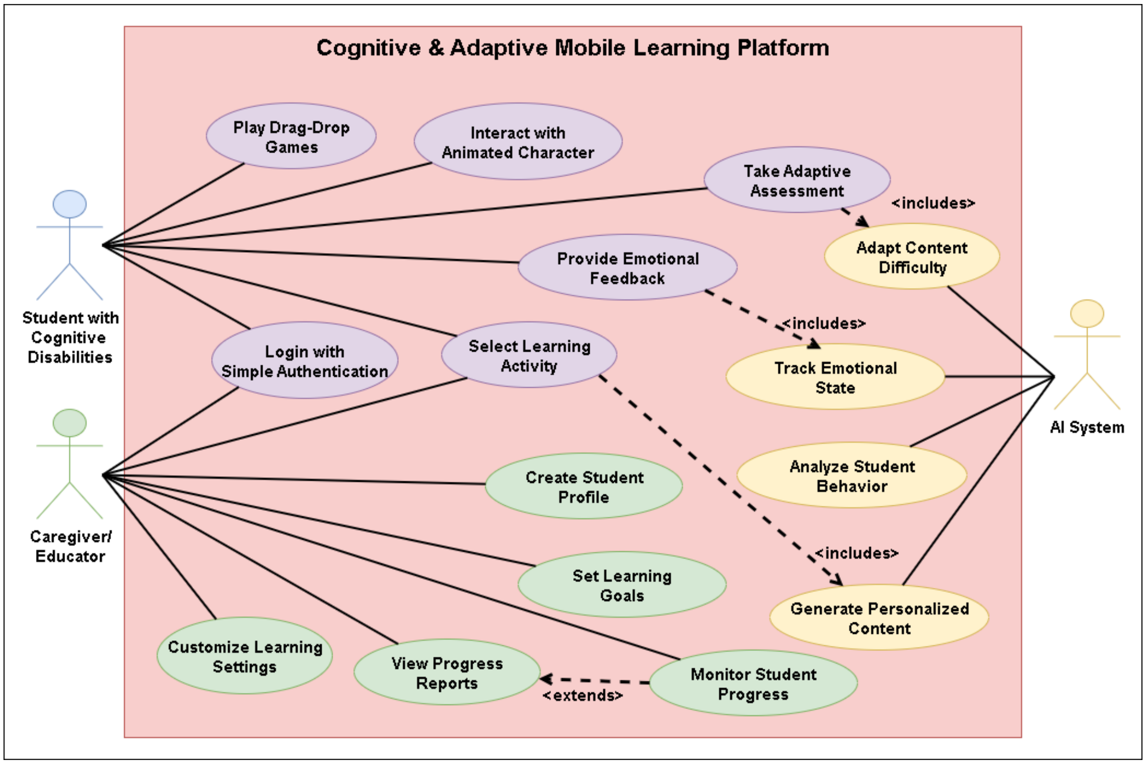

Fig. 11.1: Use case diagram for cognitive disability support

12.BUDGET & BUDGET JUSTIFICATION

Here, all costs are shown in Sri Lankan Rupees (LKR).

Table 12.1: Budget Plan for Inclusive Mobile Learning App for cognitive disability support

Purpose	Estimate cost (LKR)
Simplified content development	5000.00
Gamification Tools	10000.00
Memory and attention testing tools	5000.00
Testing and feedback sessions	5000.00
Internet facility	2000.00
Documentation and printing	3000.00
Total	30000.00

13.COMMERCIALIZATION ASPECTS OF THE PRODUCT

The proposed Cognitive & Adaptive Mobile Learning Platform has strong potential for commercialization within Sri Lanka and beyond. The solution directly addresses a critical gap in inclusive education by providing localized, curriculum-aligned, and IQ-adaptive learning content in Sinhala, supported by gamification and emotional guidance. These unique features differentiate the system from existing international tools, which are not culturally or linguistically adapted to the Sri Lankan context.

The initial market includes Sri Lankan primary schools, special education units, and non-governmental organizations that support children with disabilities. With increasing emphasis on inclusive education by the Ministry of Education, schools are seeking affordable, scalable, and digital tools to support students with cognitive and learning difficulties. The platform's cross-platform compatibility (Android/iOS), offline functionality, and low device requirements make it practical for use in rural and resource-constrained areas.

Beyond schools, the product can be extended to private tutoring centers, rehabilitation clinics, and parental use at home. Commercialization pathways include licensing to educational institutions, subscription-based access for caregivers, and partnerships with NGOs or government bodies. The modular design also allows the system to be adapted for other disability categories (visual, hearing, physical) and for other local languages in the region, increasing its long-term scalability.

The product aligns with the growing demand for digital education tools in South Asia, which has accelerated after the COVID-19 pandemic. By focusing on cultural relevance, curriculum integration, and evidence-based pedagogy, this platform can establish itself as a niche but essential solution in the educational technology market. Future commercialization strategies may also include collaboration with telecommunication providers for subsidized access and expansion to broader South Asian contexts with similar educational challenges.

14.WORK BREAKDOWN STRUCTURE

Table 14.1: Gantt chart

Time / Task	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8
Requirement analysis								
Data gathering, Presentation								
Prototype Design								
App Development								
AI related Developments								
Integration & Testing								
Validation & Feedback								
Final Presentation								

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16.APPENDICES

Appendix A: Application logo

Appendix B: Application Loading Page Interface

Appendix C: User Interface for Disability Type Selection

Appendix D: Sample activities for IQ test

Appendix E: Sample Activities

Appendix F: Activity completion and reward display

Appendix G: Data collection visit at NIE

Appendix H: Teaching and Learning materials for students with cognitive and learning difficulties

Appendix I: plagiarism Report